8. Approved Suppliers Directory & Procurement Governance

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Primary OWL Classes: `Supplier`, `Person`, `Material`, `Concrete`, `Brick`, `Steel`, `Wood`, `Insulation`, `Glass`
Primary OWL Properties: `suppliesMaterial`, `worksOn`

8.1 Modeling Suppliers in the Construction Company Ontology

In `stavebna_firma_ontology - kópia.owl`, a `Supplier` is modeled as a direct subclass of `Person` (representing an accredited supply partner or vendor liaison who collaborates on project logistics, thus inheriting `worksOn some Project` from `Person`). The class `Supplier` is governed by two formal OWL restrictions on the object property `suppliesMaterial`:

Formal OWL Axioms for `Supplier` and `suppliesMaterial`:

- **Domain & Range:** `suppliesMaterial` has domain `Supplier` and range `Material`.
- **Existential Restriction (`owl:someValuesFrom`):** `Supplier ⊑ suppliesMaterial some Material` (Every supplier must supply at least one construction material).
- **Universal Restriction (`owl:allValuesFrom`):** `Supplier ⊑ suppliesMaterial only Material` (Everything supplied via `suppliesMaterial` by a `Supplier` must belong to the class `Material`—a supplier under this property cannot supply non-material entities such as employees).
- **Concrete Specific Restriction:** The class `Concrete` additionally specifies a restriction linking `Concrete` and `Supplier` via `suppliesMaterial`, ensuring full supply-chain traceability for all structural concrete batches.

8.2 Master Directory of Approved Suppliers (2026)

ApexBuild Construction Corp. procures materials exclusively from six ISO 9001 and ISO 14001 certified supply partners, covering all six subclasses of `Material`:

Supplier ID	Supplier Name & Liaison (`Supplier`)	Supplied Ontology Material Class (`suppliesMaterial`)	Lead Time & Logistics Capacity	Key Projects Supplied
SUP-01	**Danucem ReadyMix Slovakia (Ing. Marek Holub)**	`Concrete` (C25/30 up to C60/75, waterproof & sulfate-resistant mixes)	Just-in-Time (JIT) within 45 mins from 3 batching plants (up to 180 m³/hour)	`EuroveaTower`, `FamilyHouseExample`, Danube Atrium `OfficeBuilding`
SUP-02	**Wienerberger Ceramic Systems (Ing. Zuzana Krížová)**	`Brick` (Porotherm thermal blocks, acoustic bricks, ceramic ceiling beams)	3–5 business days via flatbed `Truck` fleet	`FamilyHouseExample`, Trenčín Heritage `Reconstruction`

SUP-03	**SteelCore Europe / U.S. Steel Partner (Ing. Dávid Urban)**	`Steel` (S355 structural H-beams, castellated beams, B500B rebar coils)	14 days for standard profiles; 28 days for custom pre-cambered trusses	Danube Atrium `OfficeBuilding`, Trnava `IndustrialProject`, `EuroveaTower`
SUP-04	**Carpathian Timber & CLT Works (Ing. Lukáš Oravec)**	`Wood` (FSC-certified C24 spruce, Glulam GL24h arches, 5-ply CLT panels)	21 days CNC pre-cut delivery directly to site	Tatry CLT `WoodenHouse` Cluster, Bratislava Port Eco-Tower (`FutureProject`)
SUP-05	**IsoTherm Central Europe (Mgr. Petra Horská)**	`Insulation` (Rockwool A1 facade stone wool, XPS perimeter boards, PIR roof panels)	48 hours from regional warehouse in Senec	All `ResidentialProject`, `CommercialProject`, and Hospital `Reconstruction`
SUP-06	**Glaverbel Architectural Glazing (Ing. FilipBartoš)**	`Glass` (Triple-glazed curtain wall units, laminated safety glass, solar-control IGU)	4–6 weeks custom fabrication per facade grid elevation	`EuroveaTower`, Danube Atrium `OfficeBuilding`

8.3 Procurement SLA, Traceability & Penalty Clauses

1. **Digital Batch Traceability:** Every delivery note from a `Supplier` must contain the target `Project` ID, the ontology `Material` subclass code, and the assigned `SiteManager` signature upon unloading.
2. **Concrete Window SLA:** Because fresh `Concrete` begins hydration immediately upon mixing, supplier SUP-01 (Danucem) guarantees delivery to the project site within 60 minutes of batching. Any mixer `Truck` exceeding 90 minutes from batching time is rejected at the gate by the `SiteManager`.
3. **Sustainability & EPD Mandate:** Starting in 2026, all suppliers of `Concrete`, `Steel`, and `Wood` must provide Type III Environmental Product Declarations (EPD) compliant with EN 15804.